

25^{Years}

المدرسة الوطنية للعلوم
التطبيقية - مراكش
ECOLE NATIONALE DES SCIENCES
APPLIQUEES - MARRAKECH

ICRAMCS 2025

THE SEVENTH EDITION OF THE INTERNATIONAL CONFERENCE ON
RESEARCH IN APPLIED MATHEMATICS AND COMPUTER SCIENCE
April 24-25-26, 2025 | Marrakech, Morocco



Certificate of Attendance and Presentation

This certificate recognizes that

JAMAL ATTMANI

(in collaboration with ATLAS ABDELGHAFOR, KARAMI FAHD)

attended the seventh edition of the **International Conference on Research in Applied Mathematics and Computer Science (ICRAMCS 2025)** and presented a contribution titled:

A Novel Dynamic Threshold Approach for Image Restoration



International Conference on Research
in Applied Mathematics and Computer
Science ICRAMCS
Coordinator : Prof. Youssef EL FOUTAYENI

Prof. Youssef EL FOUTAYENI

Chairman of ICRAMCS'2025

Marrakech, Morocco, April 24-26, 2025

ICRAMCS 2025

THE SEVENTH EDITION OF THE INTERNATIONAL CONFERENCE ON
RESEARCH IN APPLIED MATHEMATICS AND COMPUTER SCIENCE

April 24-25-26, 2025 | Marrakech, Morocco



A Novel Dynamic Threshold Approach for Image Restoration

Communication Info

Authors:

Abdelghafour ATLAS¹

Jamal ATTMANI¹

Fahd KARAMI²

¹LMSC, Cadi Ayyad University,

ENSA Marrakech, Morocco

²MIMSC, Cadi Ayyad University,

EST Essaouira, Morocco

Keywords:

(1) Image Processing

(2) Perona-Malik Equation

(3) Dynamical threshold

Abstract

This study presents a mathematical and numerical analysis of a novel image denoising model. The model is developed by combining the H-1-norm decomposition approach with a dynamic threshold-based Perona-Malik (DTPM) method. By incorporating a dynamic threshold function into the edge indicator, the model adaptively adjusts the diffusion mode and strength based on the image's characteristics. Using Schauder's fixed point theorem, we prove the existence and uniqueness of weak solutions for the proposed model. Finally, numerical experiments and comparative analyses are conducted to demonstrate the model's effectiveness and performance..

© ICRAMCS 2025 Proceedings ISSN: 2605-7700

References

- [1] Li, Q., Hu, Y. and Cao, Y., 2023. Adaptive Perona-Malik model based on dynamical threshold for image multi-noise removal with details preservation. *Computers & Mathematics with Applications*, 137, pp.28-43.
- [2] L. Afraites, A. Atlas, F. Karami, D. Meskine, Some class of parabolic systems applied to image processing, *Discrete & Continuous Dynamical Systems-B* 21, no. 6 (2016): 1671.
- [3] Z. Guo, J. Yin, Q. Liu, On a reaction-diffusion system applied to image decomposition and restoration, *Math. Comput. Modelling*, 53 (2011), no. 5-6, 1336-1350.